

# Yoonhyeok Choi

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## EDUCATION

### Yonsei University

Bachelor of Science Student in Computer Science

Seoul, Republic of Korea

2022 – Ongoing (Junior)

- Relevant courses : AI Systems, Computer Architecture, Computer Vision, Data Structures, Linear Algebra, Machine Learning, Operating Systems, Probability and Statistics, Reinforcement Learning, Signal Processing for AI

## PUBLICATIONS (\*, † : Equal contributors)

1. Bo Kyu Choi\*, **Yoonhyeok Choi\***, Sooyoung Jang, Woo Seok Ha, Soomi Cho, Kimoon Jang, Beomseok Sohn, Kyung Min Kim†, Yu Rang Park†. “Multimodal Deep Learning Model for Prediction of Prognosis in Central Nervous System Inflammation”, *Brain Communications*, 2025.
2. Bo Kyu Choi, Jaehyeok Jang, **Yoonhyeok Choi**, Kyung Min Kim, Yu Rang Park. “Deep learning model for the prognostic prediction of acute inflammation of the central nervous system using T2-weighted brain magnetic resonance imaging”, *Pacific Symposium on Biocomputing*, 2025. (Poster Presentation)
3. Bo Kyu Choi, **Yoonhyeok Choi**, Yu Rang Park, Kyung Min Kim. “Development of an Artificial Intelligence Model for Diagnosis and Classification of Idiopathic Generalized Epilepsy using 3D Brain Magnetic Resonance Imaging”, *Korean Epilepsy Congress*, 2024. (Poster Presentation)

## PATENTS

1. Yu Rang Park, Bo Kyu Choi, **Yoonhyeok Choi**. A device for providing an early classification and prognosis prediction model for central nervous system inflammatory diseases based on multimodal deep learning, and its control method. Korean Patent No. 10-2024-0092215 (2024).

## EXPERIENCE

### Gangnam Severance Hospital

Department of Neurology, Yonsei University College of Medicine

Seoul, Republic of Korea

**Undergraduate Research Intern** (Advisors: Prof. Won Ju Kim, Prof. Bo Kyu Choi)

2025/11/13 – Ongoing

- Developed a thorough understanding of medical signal (EEG, PSG) data and relevant analysis methods
- Managed and processed sensitive clinical data within Severance Hospital’s restricted internal network
- Played a key role in securing a national research grant from the Korea Health Industry Development Institute (KHIDI)

### Republic of Korea Army

Compulsory military service as per the regulations of the South Korean government

Republic of Korea

**Imagery Intelligence (IMINT) Operator**

2024/05/13 – 2025/11/12

### Digital Healthcare Laboratory

Department of Biomedical Systems Informatics, Yonsei University College of Medicine

Seoul, Republic of Korea

**Undergraduate Research Intern** (Advisor: Prof. Yu Rang Park)

2023/11/01 – 2024/04/30

- Constructed and deployed public repositories on Github for academic publications utilizing the Git workflow
- Processed and analyzed medical imaging (mainly T1W MRIs) and clinical data for research applications
- Executed resource-intensive model training and optimization within remote multi-GPU Linux environments
- Gained hands-on experience in the academic paper writing process including writing, submission, review and revision

## CERTIFICATIONS

**The Medical Scientist Training Program (Undergraduate Course)** (2024/08/24)

Yonsei University

## PERSONAL INFORMATION

- **Skills:** Technical - Python, PyTorch, Linux / Languages - English (Fluent), Korean (Native)
- **Interests :** Generative models, Reinforcement Learning, AI interpretability, Medical applications of AI.
- Passionate about technical communications and pedagogy, especially the job of making complex concepts easier to understand for listeners using comprehensive visualizations and adequate metaphors.